

PRODUCT CATALOGUE · 2026 EDITION

SAP ENGINEERING WORKS

Manufacturers & Exporters of High-Precision Industrial Flanges, Pipe Fittings & Fasteners

- Stainless Steel
- Carbon Steel
- Alloy Steel

PRODUCT RANGE	Stainless Steel, Carbon Steel & Alloy Steel Industrial Components
STANDARDS	ASME / ANSI / BS / IS / DIN / EN / JIS

Our complete range spans **flanges, butt weld fittings and forged fittings** across carbon steel, mild steel, stainless steel and alloy steel — manufactured to international dimensional standards.

MATERIAL CATEGORY	GRADES & SPECIFICATIONS	TYPE / ITEMS OFFERED	DIMENSIONAL STANDARDS
Carbon Steel Flanges	ASME SA105, ASME SA516	Slip On, SW, Welding Neck, Threaded, Lap Joint, Blind Flanges	ASME B16.5, ASME B16.47, BS-10 (Table D,E,F,H,J,K,R,S,T), IS 6392, JIS B220, AWWA, EN 1092-1 (PN-10,16,25,40,63,100)
Mild Steel Flanges	IS 2062	Slip On Flanges, Blind Flanges	ASME B16.5, ASME B16.47, BS-10, IS 6392, JIS B220, AWWA, EN 1092-1
Stainless Steel Flanges	ASME SA182 304/304L/304H, 316/316L/316H, 321/321H, ASME SA240 304, 316, 321	Slip On, SW, Welding Neck, Threaded, Lap Joint, Blind Flanges	ASME B16.5, ASME B16.47, BS-10, IS 6392, JIS B220, AWWA, EN 1092-1
Alloy Steel Flanges	ASME SA182 F91, F92, F11 (CI 1,2,3), F22 (CI 1,3)	Slip On, SW, Welding Neck, Threaded, Lap Joint, Blind Flanges	ASME B16.5, ASME B16.47, BS-10, IS 6392, JIS B220, AWWA, EN 1092-1
Carbon Steel Fittings (Buttweld)	ASME SA234 WPB	90°/45° LR & SR Elbows, 90°/45° 3D Elbows, Equal/Unequal Tees, Concentric/Eccentric Reducers, Equal Cross, 180° LR/SR U Bends, End Caps	ASME B16.9
Stainless Steel Fittings (Buttweld)	ASME SA403 304 (H,L,N,LN), 316 (H,L,N,LN), 321 (H)	90°/45° LR & SR Elbows, Tees, Reducers, Crosses, 180° U Bends, End Caps	ASME B16.9
Alloy Steel Fittings (Buttweld)	ASME SA234 WP1, WP12, WP11, WP22, WP5, WP9, WP91, WP911 (CI 1,2,3)	90°/45° LR & SR Elbows, 3D Elbows, Reducers, Tees, Crosses, 180° Bends, End Caps	ASME B16.9
Forged Fittings (Carbon Steel)	ASME SA105	90°/45° Elbows, Concentric Reducers, Equal/Unequal Tees, Full/ Half Couplings (SW / Threaded)	ASME B16.11
Forged Fittings (Stainless Steel)	ASME SA182 304, 316, 321 (H, L, N, LN)	90°/45° Elbows, Reducers, Tees, Couplings (Socket Weld / Threaded)	ASME B16.11
Forged Fittings (Alloy Steel)	ASME SA182 F91, F92, F11 (CI 1,2,3), F22 (CI 1,3)	90°/45° Elbows, Reducers, Tees, Couplings (Socket Weld / Threaded)	ASME B16.11

2

FLANGE DIMENSIONS — ASME B16.5

ASME B16.5 — Class 150 Flanges (Dimensions in mm)

NPS	DN	O (OD)	FLANGE THK (TF)		HUB DIA (X)	BORE (B)			LENGTH THRU HUB (Y)			RAISED FACE		DRILLING	DRILLING
			SO/ SW/ THRD	WN		SO/ SW	WN	SO/ SW/ THRD	LJ	WN	DIA (R)	THK (TR)	PCD (W)	HOLE (D)	NO (N)
1/2"	15	90.0	9.6	11.2	30.0	22.2	22.9	16.0	16.0	48.0	34.9	2	60.3	15.9	4
3/4"	20	100.0	11.2	12.7	38.0	27.7	28.2	16.0	16.0	53.0	42.9	2	69.9	15.9	4
1"	25	110.0	12.7	14.3	49.0	34.5	34.9	18.0	17.0	56.0	50.8	2	79.4	15.9	4
1 1/4"	32	115.0	14.3	15.9	59.0	43.2	43.7	21.0	21.0	58.0	63.5	2	88.9	15.9	4
1 1/2"	40	125.0	15.9	17.5	65.0	49.5	50.0	23.0	22.0	62.0	73.0	2	98.4	15.9	4
2"	50	150.0	17.5	19.1	78.0	61.9	62.5	26.0	25.0	64.0	92.1	2	120.7	19.0	4
2 1/2"	65	180.0	20.7	22.3	90.0	74.6	75.4	29.0	29.0	70.0	104.8	2	139.7	19.0	4
3"	80	190.0	22.3	23.9	108.0	90.7	91.4	31.0	30.0	70.0	127.0	2	152.4	19.0	4
3 1/2"	90	215.0	22.3	23.9	122.0	103.4	104.1	32.0	32.0	72.0	139.7	2	177.8	19.0	8
4"	100	230.0	22.3	23.9	135.0	114.3	116.8	34.0	33.0	77.0	157.2	2	190.5	19.0	8
5"	125	255.0	22.3	23.9	164.0	143.8	144.4	37.0	36.0	89.0	185.7	2	215.9	22.2	8
6"	150	280.0	23.9	25.4	192.0	170.7	171.4	40.0	40.0	89.0	215.9	2	241.3	22.2	8
8"	200	345.0	27.0	28.6	246.0	221.5	222.2	45.0	44.0	102.0	269.9	2	298.5	22.2	8
10"	250	405.0	28.6	30.2	305.0	275.2	277.4	50.0	49.0	102.0	323.8	2	362.0	25.4	12
12"	300	485.0	30.2	31.8	365.0	327.0	328.2	56.0	56.0	115.0	381.0	2	431.8	25.4	12
14"	350	535.0	33.4	35.0	400.0	359.2	360.2	58.0	79.0	127.0	412.8	2	476.3	28.6	12
16"	400	595.0	35.0	36.6	457.0	410.5	411.2	64.0	87.0	127.0	469.9	2	539.8	28.6	16
18"	450	635.0	38.1	39.7	505.0	461.8	462.3	69.0	97.0	140.0	533.4	2	577.9	31.7	16
20"	500	700.0	41.3	42.9	559.0	513.1	514.4	73.0	103.0	145.0	584.2	2	635.0	31.7	20
24"	600	815.0	46.1	47.7	663.0	616.0	616.0	83.0	111.0	153.0	692.2	2	749.3	34.9	20

Note: "AH" refers to hub diameter for Welding Neck flanges; SO/SW bore values shown in the Bore column adjacent to the WN bore.

ASME B16.5 — Class 300 Flanges (Dimensions in mm)

NPS	DN	OD (O)	THK SO/ SW (TF)	THK WN (TF)	HUB DIA (X)	BORE WN (B)	LENGTH SO/ SW (Y)	LENGTH WN (Y)	RF DIA (R)	PCD (W)	HOLE (D)	NO (N)
1/2"	15	95.0	12.7	14.3	38.0	22.9	23.0	53.0	34.9	66.7	15.9	4
3/4"	20	115.0	14.3	15.9	48.0	28.2	26.0	58.0	42.9	82.6	19.0	4
1"	25	125.0	15.9	17.5	54.0	34.9	27.0	62.0	50.8	88.9	19.0	4
1 1/2"	40	155.0	19.1	20.7	70.0	50.0	31.0	69.0	73.0	114.3	22.2	4
2"	50	165.0	20.7	22.3	84.0	62.5	34.0	70.0	92.1	127.0	19.0	8
3"	80	210.0	27.0	28.6	117.0	91.4	43.0	80.0	127.0	168.3	22.2	8
4"	100	255.0	30.2	31.8	146.0	116.8	48.0	86.0	157.2	200.0	22.2	8
6"	150	320.0	35.0	36.6	206.0	171.4	53.0	99.0	215.9	269.9	22.2	12
8"	200	380.0	39.7	41.3	260.0	222.2	62.0	112.0	269.9	330.2	25.4	12
10"	250	445.0	46.1	47.7	321.0	277.4	67.0	118.0	323.8	387.4	28.6	16
12"	300	520.0	49.3	50.8	375.0	328.2	73.0	131.0	381.0	450.8	31.7	16

3 FORGED FITTINGS DIMENSIONS — ASME B16.11

Socket Weld Fittings — Class 3000, 6000 & 9000 (Dimensions in mm)

NOM. PIPE SIZE	SOCKET BORE DIA (B)			BORE DIA FITTING (D)	SOCKET WALL THK (C)			BODY WALL THK			CTR TO BOTTOM OF SOCKET (A)		LAYING LENGTH	
	3000	6000	9000		3000	6000	9000	3000	6000	9000	ELBOW/ TEE	45° ELBOW	COUPLING (E)	HALF COUPLING (F)
1/2"	22.2	16.6	12.5	7.2	4.67	5.97	9.35	3.73	4.78	7.47	15.50	11.00	9.50	22.50
3/4"	27.6	21.7	16.3	11.8	4.90	6.96	9.78	3.91	5.56	7.82	19.00	13.00	9.50	24.00
1"	34.3	27.4	21.5	16.0	5.69	7.92	11.38	4.55	6.35	9.09	22.50	14.00	12.50	28.50
1 1/4"	43.1	35.8	30.2	23.5	6.07	7.92	12.14	4.85	6.35	9.70	27.00	17.50	12.50	30.00
1 1/2"	49.2	41.6	34.7	28.7	6.35	8.92	12.70	5.08	7.14	10.15	32.00	20.50	12.50	32.00
2"	61.7	53.3	43.6	38.9	6.93	10.92	13.84	5.54	8.74	11.07	38.00	25.50	19.00	41.00
2 1/2"	74.4	64.2	—	—	8.76	—	—	7.01	—	—	41.00	28.50	19.00	43.00
3"	90.3	79.4	—	—	9.52	—	—	7.62	—	—	57.00	32.00	19.00	44.50

Threaded Fittings — Class 2000, 3000 & 6000 (Dimensions in mm)

NPS	CTR TO END (A) — ELBOW/TEE			CTR TO END (C) — CROSS			OD (H)			MIN WALL (G)	COUPLING (W)	CAP END (P)	MIN THRD LENGTH (L2)
	2000	3000	6000	2000	3000	6000	2000	3000	6000	3000/6000	3000/6000	3000/6000	MIN
1/2"	28	33	38	22	25	28	33	38	46	3.18	4.09	51	37/38
3/4"	33	38	44	25	28	33	38	46	56	3.18	4.32	51/60	41/43
1"	38	44	51	28	33	35	46	56	62	3.68	4.98	60	44/46
1 1/4"	44	51	60	33	35	43	56	62	75	3.89	5.28	67	44/48
1 1/2"	51	60	64	35	43	44	62	75	84	4.01	5.56	79	44/48
2"	60	64	83	43	44	52	75	84	102	4.27	7.14	86	48/51

4 BUTTWELD FITTINGS DIMENSIONS — ASME B16.9

Elbows, Tees & Crosses — Center-to-End Dimensions (mm)

NPS	PIPE OD	90° LR ELBOW (A)	45° LR ELBOW (B)	90° SR ELBOW (C)	90° 3D ELBOW (D)	45° 3D ELBOW (E)	EQUAL TEE & CROSS			180° LR RETURN
							RUN (F)	OUTLET (G)	CENTER (H)	BACK-TO- FACE (J)
1/2"	21.3	38	16	—	—	25	25	25	76	56
3/4"	26.7	38	19	—	57	24	29	29	76	51
1"	33.4	38	22	25	76	31	38	38	76	56
1 1/2"	48.3	57	29	38	114	47	57	57	114	83
2"	60.3	76	35	51	152	63	64	64	152	106
2 1/2"	73.0	95	44	64	190	79	76	76	190	132
3"	88.9	114	51	76	229	95	86	86	229	159
4"	114.3	152	64	102	305	127	105	105	305	210
6"	168.3	229	95	152	457	189	143	143	457	313
8"	219.1	305	127	203	610	252	178	178	610	414
10"	273.0	381	159	254	762	316	216	216	762	518
12"	323.8	457	190	305	914	378	254	254	914	619

Concentric & Eccentric Reducers — ASME B16.9 (mm)

NOMINAL PIPE SIZE	OD LARGE END	OD SMALL END	END TO END (N)	NOMINAL PIPE SIZE	OD LARGE END	OD SMALL END	END TO END (N)
3/4" x 1/2"	26.7	21.3	38	4" x 3"	114.3	88.9	102
1" x 3/4"	33.4	26.7	51	4" x 2"	114.3	60.3	102
1 1/2" x 1"	48.3	33.4	64	6" x 4"	168.3	114.3	140
2" x 1 1/2"	60.3	48.3	76	8" x 6"	219.1	168.3	152
2" x 1"	60.3	33.4	76	10" x 8"	273.0	219.1	178
3" x 2"	88.9	60.3	89	12" x 10"	323.8	273.0	203

5 PIPE SCHEDULE WALL THICKNESS CHART

Per **ANSI B36.10 / B36.19**. All dimensions in millimeters (mm), per standard ASME pipe schedules.

PIPE SIZE	OD (MM)	SCH 5S	SCH 10S	SCH 20	STD	SCH 40	XS	SCH 80	SCH 120	SCH 160	XXS
1/2"	21.33	1.65	2.11	—	2.77	2.77	3.73	3.73	—	4.78	7.47
3/4"	26.67	1.65	2.11	—	2.87	2.87	3.91	3.91	—	5.56	7.82
1"	33.40	1.65	2.77	—	3.38	3.38	4.55	4.55	—	6.35	9.09
1 1/4"	42.16	1.65	2.77	—	3.56	3.56	4.85	4.85	—	6.35	9.70
1 1/2"	48.26	1.65	2.77	—	3.68	3.68	5.08	5.08	—	7.14	10.15
2"	60.33	1.65	2.77	—	3.91	3.91	5.54	5.54	—	8.74	11.07
2 1/2"	73.03	2.11	3.05	—	5.16	5.16	7.01	7.01	—	9.52	14.02
3"	88.90	2.11	3.05	—	5.49	5.49	7.62	7.62	—	11.13	15.24
4"	114.30	2.11	3.05	—	6.02	6.02	8.56	8.56	11.13	13.49	17.12
6"	168.28	2.77	3.40	—	7.11	7.11	10.97	10.97	14.27	18.26	21.95
8"	219.08	2.77	3.76	6.35	8.18	8.18	12.70	12.70	18.26	23.01	22.23
10"	273.05	3.40	4.19	6.35	9.27	9.27	12.70	15.09	21.44	28.58	25.40
12"	323.85	3.96	4.57	6.35	9.52	10.31	12.70	17.48	25.40	33.32	25.40

QUALITY POLICY & CERTIFICATION

- 100% Raw material test certificates provided as per EN 10204 3.1 / 3.2
- Hydrostatic testing, PMI testing, Ultrasonic testing, and Radiography available upon request
- Custom machining, beveling, and specific pressure ratings executed precisely to client drawings

HOW TO ORDER / REQUEST QUOTES

- Material Grade
- Nominal Pipe Size (NPS)
- Class / Schedule
- Standard (e.g. ASME B16.5 / B16.9)
- Quantity required

Thank you for considering SAP Engineering Works

For quotations, technical drawings, or custom requirements, please reach out with your specifications.